

Predicting TCR Cross-Reactivity Before It Reaches a Patient

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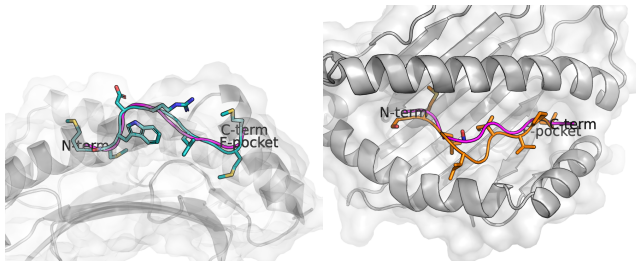


Figure 1: **Proposed framework.** Diffusion models sample from the cognate conformation (teal, left) toward a known off-target one (orange, right). Each generated backbone (magenta) is scored against both.

The problem. TCR-engineering therapies promise a transformative future for cancer treatment, and the first reached approval in 2024¹, yet cross-reactivity remains a fundamental barrier to scaling them into personalized medicine. The danger is not hypothetical. An affinity-enhanced MAGE-A3 receptor that had cleared extensive preclinical testing caused fatal cardiac toxicity in two patients², cross-recognizing an unrelated self-peptide from the muscle protein titin that was identified only after the fact³.

Why the field is stuck. The bottleneck has never been imagination, it has been throughput. Experimental validation is slow and costly, an untenable price when the presented peptide space is astronomically large, on the order of 10^{11} , and the fast-growing tumors these therapies target leave only a narrow clinical window to act. The extreme polymorphism of HLA leaves binding data sparse for most alleles and for underrepresented populations, precisely where personalized therapy is needed most. Amino-acid scanning resolves one receptor at a time and does not scale to the combinatorics of the peptide–TCR interaction.

The approach. The tools to test this at scale have only now come within reach. I would bring two complementary tools of my own design, and pair them with inverse folding as the third component. ESMCBA is a binding-affinity system built on protein language models with domain adaptation that recovers usable signal even for data-scarce MHC alleles⁴. A structure-guided generative framework uses diffusion models conditioned on the MHC groove to sample compatible peptides across high-priority alleles⁵. Inverse-folding models, which read a fixed backbone and decode the sequences its geometry admits, supply what neither of the first two can, namely which peptide positions a given re-

ceptor actually reads⁶. Together they let me computationally screen these interactions and chart the cross-reactive neighborhood of a receptor before it ever reaches a patient.

How it is proven. Each prediction is falsifiable at the bench. The receptor either engages the flagged peptide or it does not. Three experiments establish that, in order.

Experiment 1, define the search axes. Inverse-folding models will be applied to a panel of solved pMHC–TCR structures twice, once with the receptor present and once with it removed. Positions whose native recovery collapses on removal are the positions that receptor reads, and they define the axes along which a cross-reactive peptide may differ. Molecular-dynamics ensembles of each complex, rather than the single crystal snapshot, will set how much conformational variation still counts as the same presentation, so the tolerance is measured rather than assumed. Predicted read positions will be checked against measured mutations from SKEMPI and IEDB matched to each structure’s own receptor by CDR3. An architecture-matched model retrained without MHC or TCR structures serves as the control separating learned interface chemistry from memorized training data.

Experiment 2, retrospective recovery. The full screen will be run blind on receptors whose off-targets are already documented. MAGE-A3 is the primary test, since the titin epitope is known and was missed by the methods in use at the time. A screen that fails to rank it fails outright, and this is the pass condition for proceeding.

Experiment 3, prospective validation. Top-ranked candidates for a panel of characterized receptors go to binding measurement, and those that bind go to functional assay. The primary deliverable is a false-negative rate for cross-reactivity prediction on peptides never seen in training, a quantity the field currently lacks. The loop closes within the appointment.

What becomes possible. A screen that runs in silico before infusion turns cross-reactivity from a lethal surprise into a predictable and testable step, applied to every candidate receptor rather than the few that fit an experimental budget, and making the next generation of cancer therapies measurably safer. It generalizes to any therapy built on engineered molecular recognition, where the failure mode is likewise specificity, not potency. The societal stake is measured in patients, not benchmarks.

Trajectory. The first problem is chosen because it is winnable inside a year, the second because it is worth a career. Near term the deliverable is the validated screen and its false-negative rate. Beyond it lies whether specificity can be designed at the outset, not audited after.

Recent unpublished work of my own indicates both routes are tractable on real pMHC–TCR structures, which is what makes this worth building now.

¹ Singh, Mol. Ther. 32:3195, 2024.

² Linette et al., Cardiovascular toxicity and titin cross-reactivity of affinity-enhanced T cells. Blood 122:863–871, 2013.

³ Cameron et al., A titin-derived HLA-A1 peptide as cross-reactive target for MAGE-A3 T cells. Sci. Transl. Med. 5:197ra103, 2013.

⁴ Mares, Espinoza, Ioannidis, Domain-specific pre-training of protein language models for pMHC-I binding. PMLR 311:304–325, 2025.

⁵ Mares, Espinoza, Ioannidis, Structure-guided pMHC-I libraries using diffusion models. ICML GenAI Biology Wksp, 2025.

⁶ Ribeiro-Filho et al., Structure-based deep learning for TCR design. PLOS Comput. Biol., 2024.